

M.Sc. Thesis Proposal

A Human–AI Centric Usability Framework for Evaluating LLM Conversational Interfaces

1. Introduction

In recent years, AI chat systems such as ChatGPT, Gemini, and similar tools have become very popular. These systems allow users to interact with AI through a chat interface. Even though these systems use almost similar AI technology, their user interfaces (UI) and interaction behaviors are often different.

For example:

- some systems allow editing previous messages easily, others do not
- regenerate or retry actions behave differently
- system status (loading, generating, error) is shown differently
- buttons and controls are placed in different locations

Because of these differences, users often feel confused when switching between platforms.

Current usability standards, such as ISO 9241-11 usability standard, are general and not designed specifically for AI chat systems.

This research aims to develop a usability engineering framework specifically for LLM chat interfaces, along with universal design rules to make these systems more consistent and easier to use.

2. Problem Statement

LLM-based chat systems are rapidly growing, but their design and interaction patterns are not standardized. Different platforms implement features such as input interaction, response generation, feedback mechanisms, and conversation control in different ways.

These inconsistencies create several usability problems. Users often experience confusion when switching between platforms, struggle to predict system behavior, and face difficulties in understanding how different features work. As a result, user experience becomes inconsistent and less efficient.

Although general usability principles exist, they are not specifically designed for LLM-based conversational systems and do not address the unique interaction patterns of Human–AI interfaces. Furthermore, there is no structured framework to systematically evaluate and compare the usability of these systems.

Therefore, there is a need for a dedicated usability engineering framework that can evaluate, standardize, and improve the frontend usability of LLM-based chat systems.

3. Research Gap

Although usability engineering has been widely studied, there is no dedicated framework specifically designed for evaluating the usability of LLM-based chat systems. Existing usability frameworks are general in nature and were not developed for conversational Human–AI interaction.

These frameworks do not adequately address key aspects of LLM chat interfaces, such as prompt-based interaction, response regeneration, conversation control, and system feedback. In addition, they do not consider the need for consistency across multiple LLM platforms, which is essential for reducing user confusion.

Furthermore, there is no standardized set of usability guidelines or evaluation methods to systematically compare and improve the frontend interaction of these systems. Therefore, a clear research gap exists in developing a Human–AI centric usability framework that focuses on frontend interaction and supports standardization across LLM-based chat systems.

4. Aim and Objectives

Aim

To develop a universal usability engineering framework for evaluating and improving the usability of LLM-based chat systems.

Objectives

1. Study existing LLM chat systems and analyze their UI differences
2. Identify common usability problems in these systems
3. Conduct user studies to understand confusion and interaction issues
4. Design a usability evaluation framework specifically for LLM chat interfaces
5. Define a set of universal design rules for LLM chat systems
6. Validate the framework using real user testing

5. Research Questions

1. What usability problems do users face when using different LLM chat systems?
2. Which interface differences cause the most confusion?
3. How can usability of LLM chat interfaces be systematically evaluated?
4. What design rules can make LLM chat systems more consistent and predictable?
5. Can a universal framework improve user experience across different platforms?

6. Literature Review

Usability engineering is traditionally based on concepts such as effectiveness, efficiency, and user satisfaction, as defined in the ISO 9241-11 usability standard. In addition, usability evaluation methods such as heuristic evaluation proposed by Jakob Nielsen have been widely used to assess user interfaces.

These approaches have been effective for traditional software systems, where user interactions are relatively stable and predictable. However, LLM-based chat systems introduce a new form of interaction based on conversational input, dynamic response generation, and continuous system behavior.

Existing usability frameworks are not specifically designed for such systems. They do not fully consider key aspects of LLM interactions, such as prompt-based communication, response regeneration, conversation control, and system feedback. Moreover, they do not address the variability and unpredictability of AI-driven responses.

Recent research in Human–AI interaction has explored user trust, explainability, and interaction design in AI systems. However, these studies mainly focus on system behavior and user perception, rather than providing a structured usability framework for interaction in LLM chat systems. Therefore, there is a need for a dedicated usability engineering framework that focuses on frontend interaction and supports consistent and standardized design across LLM-based chat platforms.

7. Proposed Framework: Human–AI Centric Usability Framework (HAIUF)

This research proposes a Human–AI Centric Usability Framework to evaluate and standardize the usability of LLM-based chat systems.

The framework focuses on two layers:

1. Core Interaction Components
2. Usability Dimensions

1.0 Core Interaction Components

These are the main features of LLM chat systems that will be evaluated.

i. Input Box (Prompt Interface)

Focus:

- clarity of input field
- support for multiline input
- Support for adding multiple attachment
- visibility of send action

Evaluation:

- Can users easily understand how to give input?
- Can they edit and resend prompts without confusion?

ii. Answer Generation (AI Response Area)

Focus:

- response structure and formatting
- readability
- separation between user input and AI output

Evaluation:

- Is the response easy to read and understand?
- Is important information clearly structured?

iii. Regenerate Response

Focus:

- visibility of regenerate button
- behavior consistency
- clarity of outcome

Evaluation:

- Do users understand what “regenerate” does?
- Does it replace or add a new response clearly?

iv. Like / Dislike Response

Focus:

- visibility of feedback controls
- system reaction after feedback

Evaluation:

- Do users know what happens after clicking like/dislike?
- Is feedback effect communicated clearly?

v. Share Chat

Focus:

- ease of sharing conversation
- privacy clarity

Evaluation:

- Is it clear what will be shared?
- Is user control over shared content sufficient?

vi. Archive / Delete Chat

Focus:

- clarity of actions
- risk of accidental deletion

Evaluation:

- Can users safely manage conversations?
- Is there confirmation or recovery option?

vii. Usability Features

Includes:

- read aloud
- keyboard accessibility
- dictation (voice input)

Evaluation:

- Are these features visible and usable?
- Do they work consistently across devices?

viii. Add Files (Attachment Support)

Focus:

- file upload clarity
- supported formats
- feedback after upload

Evaluation:

- Is it easy to attach files?
- Is system behavior after upload clear?

ix. Web Search & Citation

Focus:

- visibility of sources
- clarity of citation

Evaluation:

- Are sources clearly shown?
- Can users trust and verify information easily?

x. Changing Model

Focus:

- model selection UI
- clarity of differences between models

Evaluation:

- Do users understand what changing the model means?
- Is system behavior predictable after switching?

2.0 Usability Dimensions

These dimensions are used to evaluate each component.

i. Response Usability

- readability and structure of responses
- ease of copying or reusing output

ii. System Feedback

- visibility of loading, generating, and error states
- clarity of system status

iii. Conversation Control

- editing prompts
- regenerating responses
- managing conversation

iv. Consistency

- same actions behave similarly across systems
- consistent placement of controls

v. Transparency

- clear explanation of system actions
- visibility of limitations and changes

vi. Accessibility

- keyboard support
- screen reader compatibility
- inclusive interaction design

8. Methodology

Phase 1: Comparative Analysis

A comparative analysis will be conducted on multiple LLM-based chat systems (e.g., ChatGPT, Gemini, and others) to examine differences in frontend usability. The analysis will focus on key interaction components such as input interface, response generation, regeneration controls, feedback mechanisms (like/dislike), conversation management (share, archive, delete), file attachment, citation display, model switching, and accessibility features.

Each system will be evaluated against defined usability dimensions, including response usability, system feedback, conversation control, consistency, transparency, and accessibility.

A structured comparison matrix will be used to identify similarities, differences, and usability gaps across platforms.

Phase 2: User Study

Participants will be asked to perform tasks such as:

- entering and submitting a prompt using the input box
- editing and resubmitting a previous prompt
- regenerating a response and comparing outputs
- providing feedback using like/dislike controls
- sharing a conversation using the share feature
- archiving or deleting a conversation
- attaching a file and interacting with the response
- identifying and reviewing web search citations
- switching between different AI models
- using accessibility features such as keyboard navigation or read-aloud

Data will be collected:

Raw Data (Collected)

- Time
- Errors
- Confusion
- Feedback

Derived Metrics (Calculated)

- Task Success Rate
- Efficiency Score
- Cognitive Load Indicator
- Overall Usability Score

Data will be collected through task-based usability testing. Participants will be asked to perform predefined tasks while their interactions are recorded. The time taken to complete each task will be measured using screen recordings. Errors will be counted based on incorrect actions or failed attempts during task execution. Confusion points will be identified through observation of hesitation, repeated actions, and user comments during the session. User feedback will be collected using short questionnaires and post-task interviews.

Phase 3: Comparison Matrix

Table I. Component Comparison Table

Component	Evaluation Criteria	ChatGPT	Gemini	Tool X	Notes
Input Box	Clear input field, multiline support, send visibility	✓ / ✗ / Partial			
Answer Generation	Structured, readable, well-formatted response				

Regenerate Response	Clear function, predictable behavior				
Like/Dislike	Visible, understandable effect				
Share Chat	Easy to use, clear sharing scope				
Archive/Delete	Safe action, confirmation available				
Usability Features	Read aloud, keyboard, dictation				
Add Files	Easy upload, clear feedback				
Web Citation	Clear sources, clickable references				
Model Switching	Clear options, predictable behavior				

Table II. Usability Dimension Matrix

Component	Dimension	ChatGPT	Gemini	Tool X	Remarks
Input Box	Accessibility				
Input Box	Consistency				
Answer	Response Usability				
Answer	Transparency				
Regenerate	System Feedback				
Regenerate	Consistency				
Share Chat	Transparency				
Delete Chat	Error Prevention				

Table III. Scoring Matrix

Component	Dimension	ChatGPT	Gemini	Tool X
Input Box	Accessibility	4	3	
Input Box	Consistency	5	3	
Regenerate	Clarity	4	2	
Share Chat	Transparency	3	2	

Scoring Guide:

Meaning	Not available	Very poor	Poor	Average	Good	Excellent
Score	0	1	2	3	4	5

Table IV. Gap Identification Table

Component	Issue Identified	Affected Systems	Usability Impact
Regenerate	Behavior inconsistent	Gemini	User confusion
Model Switch	Not clearly explained	Multiple	Low transparency
Share Chat	Scope unclear	Tool X	Privacy concern

Phase 4: Framework Development

Based on the findings from comparative analysis and user studies, the proposed framework will be developed through the following steps:

- define usability dimensions based on observed usability issues and existing usability principles
- map each interaction component (e.g., input box, regenerate response, share chat) to relevant usability dimensions
- develop clear evaluation criteria for each component–dimension pair
- design a standardized scoring system to measure usability
- validate the framework through expert review and user testing

Phase 5: Rule Definition

Based on the findings from comparative analysis and usability evaluation, a set of universal usability rules for LLM-based chat systems will be developed. These rules aim to improve consistency, predictability, and user experience across different platforms.

The proposed rules include:

- Consistent Placement of Controls: Core actions such as send, regenerate, and share should be placed in consistent and predictable locations.
- Clear System Feedback: The system must clearly indicate its current state, including loading, response generation, and errors.
- Predictable Regenerate Behavior: Regenerating responses should produce a clearly distinguishable result without confusing replacement or duplication.
- Editable Prompt Interaction: Users should be able to easily edit and resubmit previous prompts without losing context.

- **Transparent Feedback Mechanism:** Actions such as like/dislike should clearly communicate their purpose and system effect.
- **Safe Conversation Management:** Actions such as delete or archive should include confirmation and recovery options from archive to prevent accidental loss.
- **Visible Source and Citation Information:** When external information is used, sources should be clearly displayed and accessible.
- **Clear Model Switching Behavior:** Changing AI models should be understandable and should clearly indicate differences in behavior or output.
- **Accessible Interaction Support:** All key features should support keyboard navigation, screen readers, and voice interaction.
- **Consistent Interaction Behavior Across Platforms:** Similar actions should behave consistently across different LLM systems to reduce user confusion.

Phase 6: Validation

The proposed framework will be validated through multiple methods to ensure its reliability and effectiveness.

- The framework will be applied to multiple LLM-based chat systems (e.g., ChatGPT, Gemini) to evaluate their usability using the defined scoring system.
- The results will be compared to identify usability differences and consistency across systems.
- A prototype interface designed based on the proposed usability rules will be developed and tested to evaluate improvements in user experience. The prototype will be a functional but simplified web-based interface, focusing only on frontend interaction and usability, not backend AI complexity.
- User testing will be conducted to compare task performance, error rates, and user satisfaction between existing systems and the improved prototype.
- Expert evaluation will be performed to assess the validity and completeness of the framework.

9. Expected Contributions

This research is expected to make the following contributions:

1. A Human–AI centric usability evaluation framework specifically designed for interaction in LLM-based chat systems
2. A structured evaluation model, including usability dimensions, component–dimension mapping, and a standardized scoring system
3. A set of universal usability rules to improve consistency, predictability, and user experience across LLM platforms
4. A comparative analysis of existing LLM chat systems, identifying key usability gaps and inconsistencies
5. A validated usability assessment approach, supported by user testing and expert evaluation

6. Practical design guidelines for developers and designers to build more usable and consistent AI chat interfaces
7. A contribution to the field of Human–AI interaction by focusing on usability and interaction standardization rather than AI model performance

10. Significance of the Study

This study is important because LLM-based chat systems are increasingly used in everyday tasks such as content creation, learning, and decision support. Users often interact with multiple platforms, but differences in design and interaction behavior can create confusion and reduce usability.

Currently, there is no standardized usability framework specifically designed for LLM chat interfaces. Existing usability principles are general and do not fully address the unique interaction patterns of Human–AI systems.

This research will provide a structured approach to evaluate and improve usability in LLM-based systems. By defining usability dimensions, evaluation methods, and universal design rules, the study aims to make AI interfaces more consistent, predictable, and user-friendly.

The findings of this research will benefit both academia and industry by contributing to Human–AI interaction research and providing practical guidelines for designing better AI-driven interfaces.

11. Timeline

Phase	Duration
Literature Review	1 months
System Analysis	2 months
User Study	2 months
Framework Design	1 months
Validation & Writing	2 months

12. Conclusion

This research focuses on improving the usability of LLM chat systems by developing a standardized usability framework. It does not evaluate AI intelligence, but instead focuses on how users interact with AI systems. By defining clear usability rules and evaluation methods, this study aims to make LLM interfaces more consistent and easier to use across platforms.